

PolyFlop8 I/O Unit

Unit Test Card

Comprehensive Verification of Input/Output Operations

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Test Engineer:	Artem Horiunov
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1 TEST OVERVIEW

1.1 Unit Under Test (UUT)

Component	I/O Unit (Input/Output Controller)
VHDL Entity	IO_Unit
Source File	Processor.txt (IO_Unit entity)
Test Environment	Vivado ML 2025
IO Ports	Port A (Output), Port B (Input)
Address Width	8-bit (256 possible port addresses)
Data Width	8-bit
Operation Type	Synchronous Write, Combinational Read

1.2 Scope

This test card defines the verification procedure for the PolyFlop8 I/O Unit component. The I/O Unit provides memory-mapped I/O capabilities, allowing the CPU to interact with external devices through output ports and input pins.

1.3 I/O Port Address Map (Based on VHDL)

Address	Type	Port Name	Function
0x01	Output	Port A	Write: Sets output value for LEDs/displays Read: Returns current output value
0x02	Input	Port B	Read: Returns input pin values (buttons/switches)
Other	-	-	Read: Returns 0x00

Table 1: I/O Address Map

1.4 Signal Nomenclature (Matching VHDL)

Signal Name	Direction	Description
io_we	Input	Write Enable (1 = Write to output port)
io_addr_in	Input	8-bit I/O port address
io_data_out	Input	Data to write to output port (from CPU registers)
io_data_in	Output	Data read from I/O port (to CPU registers)
port_out_a	Output	Physical output pins (e.g., LEDs)
pin_in_b	Input	Physical input pins (e.g., buttons)

1.5 Test Objectives

- Verify OUT instruction writes register value to output port when io_we='1'

- Validate correct output port selection based on io_addr_in
- Verify IN instruction reads external pin values into register via io_data_in
- Test output port retains last written value until updated or reset
- Verify reset clears all output port registers to zero
- Test undefined address behavior for both read and write operations
- Verify timing characteristics of write operations

2 REQUIREMENTS COVERAGE

Req ID	Requirement Description	Test Case ID	Status
TR-IO-01	Verify OUT instruction writes register value to output port when io_we = '1'	TC-IO-001-01	TBD
TR-IO-02	Verify correct output port selection based on io_addr_in	TC-IO-001-02	TBD
TR-IO-03	Verify IN instruction reads external pin values into register via io_data_in	TC-IO-001-03	TBD
TR-IO-04	Verify output port retains last written value until updated or reset	TC-IO-001-04	TBD
TR-IO-05	Verify reset clears all output port registers to zero	TC-IO-001-05	TBD

Table 2: Requirements Coverage Matrix

3 DETAILED TEST CASES

3.1 Test Case TC-IO-001-01: OUT Instruction

Requirement: TR-IO-01

Purpose: Verify OUT instruction writes register value to output port when io_we = '1'.

TC ID	Test Description	Inputs/Stimuli	Expected Outputs
1.1	Basic OUT to Port A	io_addr_in = 0x01 io_data_out = 0xAA io_we = '1' clock ↑	port_out_a = 0xAA Written on clock edge
1.2	OUT with io_we='0' (no write)	io_addr_in = 0x01 io_data_out = 0x55 io_we = '0' clock ↑	port_out_a unchanged No write occurs
1.3	OUT on falling edge (no write)	Setup as 1.1 clock ↓	No write occurs Only rising edge triggers
1.4	Multiple OUT operations	Sequence: OUT 0x01, 0x11 OUT 0x01, 0x22 OUT 0x01, 0x33	Port A shows: 0x11 → 0x22 → 0x33

TC ID	Test Description	Inputs/Stimuli	Expected Outputs
1.5	OUT with different data values	Test data patterns: 0x00, 0xFF, 0x55, 0xAA, 0x0F, 0xF0, 0x99, 0x66	Each value correctly written to Port A

3.2 Test Case TC-IO-001-02: Port Address Selection

Requirement: TR-IO-02

Purpose: Verify correct output port selection based on io_addr_in.

TC ID	Test Description	Inputs/Stimuli	Expected Outputs
2.1	Write to Port A (address 0x01)	io_addr_in = 0x01 io_data_out = 0xAA io_we = '1' clock ↑	port_out_a = 0xAA Write successful
2.2	Write to undefined address (0x00)	io_addr_in = 0x00 io_data_out = 0x55 io_we = '1' clock ↑	No effect on Port A (no write to any port)
2.3	Write to undefined address (0x03)	io_addr_in = 0x03 io_data_out = 0x77 io_we = '1' clock ↑	No effect on Port A Address not mapped
2.4	Write to address 0xFF (max)	io_addr_in = 0xFF io_data_out = 0x99 io_we = '1' clock ↑	No effect on Port A Address not mapped
2.5	Address change during write cycle	io_addr_in = 0x01 Change to 0x02 just before clock edge	Write uses address at clock edge
2.6	Rapid address switching	Switch between 0x01 and other addresses during writes	Only writes to 0x01 affect Port A

3.3 Test Case TC-IO-001-03: IN Instruction

Requirement: TR-IO-03

Purpose: Verify IN instruction reads external pin values into register via io_data_in.

TC ID	Test Description	Inputs/Stimuli	Expected Outputs
3.1	IN from Port B (address 0x02)	io_addr_in = 0x02 pin_in_b = 0x55	io_data_in = 0x55 Immediate (combinational)
3.2	IN from Port A (address 0x01)	1. OUT 0xAA to Port A 2. io_addr_in = 0x01	io_data_in = 0xAA Reads back output value
3.3	IN from undefined address (0x00)	io_addr_in = 0x00	io_data_in = 0x00 Default read value
3.4	IN with changing input pins	io_addr_in = 0x02 pin_in_b = 0x00 → 0xFF	io_data_in follows immediately
3.5	IN with clock running	Run clock at 100MHz Read Port B continuously	Reads are immediate no clock dependency

TC ID	Test Description	Inputs/Stimuli	Expected Outputs
3.6	IN all-zero input pattern	pin_in_b = 0x00 io_addr_in = 0x02	io_data_in = 0x00
3.7	IN all-ones input pattern	pin_in_b = 0xFF io_addr_in = 0x02	io_data_in = 0xFF
3.8	IN alternating pattern (0x55)	pin_in_b = 0x55 io_addr_in = 0x02	io_data_in = 0x55

3.4 Test Case TC-IO-001-04: Output Retention

Requirement: TR-IO-04

Purpose: Verify output port retains last written value until updated or reset.

TC ID	Test Description	Inputs/Stimuli	Expected Outputs
4.1	Write and verify retention	1. Write 0xAA to Port A 2. Wait 100 clock cycles 3. No further writes	port_out_a = 0xAA throughout
4.2	Retention during IN operations	1. Write 0x55 to Port A 2. Perform multiple IN reads 3. No OUT operations	Port A remains 0x55 Reads don't affect output
4.3	Retention after address changes	1. Write 0x33 to Port A 2. Change io_addr_in 3. No writes with io_we='1'	Port A remains 0x33 Address change irrelevant
4.4	Retention during input changes	1. Write 0x66 to Port A 2. Change pin_in_b values	Port A remains 0x66 Input changes don't affect output
4.5	Overwrite with new value	1. Write 0x11 to Port A 2. Wait 3. Write 0x22 to Port A	Port A: 0x11 → 0x22 New value replaces old
4.6	Long-term retention test	Write value, run 1000 clock cycles Verify unchanged	Value retained throughout test

3.5 Test Case TC-IO-001-05: Reset Behavior

Requirement: TR-IO-05

Purpose: Verify reset clears all output port registers to zero.

TC ID	Test Description	Inputs/Stimuli	Expected Outputs
5.1	Reset after write	1. Write 0xAA to Port A 2. Assert reset (rst='1') 3. Deassert reset	port_out_a = 0x00 after reset
5.2	Reset during write operation	Setup for write Assert reset before clock edge	Reset takes precedence Port A = 0x00
5.3	Reset pulse width test	Assert reset for 1, 2, 10 clock cycles	Port A cleared in all cases
5.4	Reset then write	1. Reset 2. Write 0x55 to Port A	Port A: 0x00 → 0x55 Reset doesn't prevent writes
5.5	Multiple reset cycles	Write, reset, write, reset Multiple times	Each reset clears output

TC ID	Test Description	Inputs/Stimuli	Expected Outputs
5.6	Reset with active inputs	pin_in_b = 0xFF Assert reset	Port A = 0x00 Inputs unaffected

3.6 Test Case TC-IO-001-06: Timing and Integration

Purpose: Test timing characteristics and integration behavior.

TC ID	Test Description	Inputs/Stimuli	Expected Outputs
6.1	Write setup/hold times	Change io_data_out just before/after clock edge	Write uses value at clock edge
6.2	Read propagation delay	Change io_addr_in or pin_in_b, measure delay	io_data_in updates within 5ns (combinational)
6.3	Simultaneous read/write	Read from Port A while writing to it	Read sees old value until next clock edge
6.4	Back-to-back operations	IN → OUT → IN sequence in consecutive cycles	Each operation completes correctly
6.5	Maximum clock frequency	Test writes at increasing clock frequencies	Operation correct up to design limit
6.6	Power-on behavior	Simulate power-on without explicit reset	Outputs should initialize to known state

4 TEST BENCH ARCHITECTURE

4.1 VHDL Testbench Structure

```
-- IO_Unit Testbench (io_unit_tb.vhd)
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

entity io_unit_tb is
end io_unit_tb;

architecture Behavioral of io_unit_tb is
    -- Component Declaration (matches exact VHDL from Processor.txt)
    component IO_Unit
        Port (
            clk          : in  STD_LOGIC;
            reset        : in  STD_LOGIC;
            io_we         : in  STD_LOGIC;
            io_addr_in    : in  STD_LOGIC_VECTOR (7 downto 0);
            io_data_out   : in  STD_LOGIC_VECTOR (7 downto 0);
            io_data_in    : out STD_LOGIC_VECTOR (7 downto 0);
            port_out_a    : out STD_LOGIC_VECTOR (7 downto 0);
            pin_in_b      : in  STD_LOGIC_VECTOR (7 downto 0)
        );
    end component;

    -- Test signals
    signal clk_tb          : STD_LOGIC := '0';
    signal reset_tb        : STD_LOGIC := '0';
```

```

signal io_we_tb      : STD_LOGIC := '0';
signal io_addr_in_tb : STD_LOGIC_VECTOR(7 downto 0) := (others => '0');
signal io_data_out_tb : STD_LOGIC_VECTOR(7 downto 0) := (others => '0');
signal io_data_in_tb  : STD_LOGIC_VECTOR(7 downto 0);
signal port_out_a_tb  : STD_LOGIC_VECTOR(7 downto 0);
signal pin_in_b_tb    : STD_LOGIC_VECTOR(7 downto 0) := (others => '0');

constant CLK_PERIOD : time := 10 ns;

begin
  -- Unit Under Test Instantiation
  UUT: IO_Unit port map (
    clk      => clk_tb,
    reset    => reset_tb,
    io_we     => io_we_tb,
    io_addr_in => io_addr_in_tb,
    io_data_out => io_data_out_tb,
    io_data_in  => io_data_in_tb,
    port_out_a  => port_out_a_tb,
    pin_in_b    => pin_in_b_tb
  );

  -- Clock generation process
  clk_process: process
  begin
    clk_tb <= '0';
    wait for CLK_PERIOD/2;
    clk_tb <= '1';
    wait for CLK_PERIOD/2;
  end process;

  -- Main test stimulus process
  stim_proc: process
  begin
    wait for 100 ns;
    report "Starting I/O Unit Test Bench";

    -- Initial reset
    reset_tb <= '1';
    wait for 20 ns;
    reset_tb <= '0';
    wait for 20 ns;

    -- Test Case 1.1: Basic OUT to Port A
    report "Test Case 1.1: Basic OUT to Port A";
    io_addr_in_tb <= x"01";
    io_data_out_tb <= x"AA";
    io_we_tb <= '1';
    wait until rising_edge(clk_tb);
    io_we_tb <= '0';
    wait for 5 ns;
    assert port_out_a_tb = x"AA"
      report "OUT failed: Expected port_out_a = 0xAA, got " &
        to_hstring(port_out_a_tb) severity error;

    -- Test Case 1.2: OUT with io_we='0' (no write)
    report "Test Case 1.2: OUT with io_we='0'";
    io_addr_in_tb <= x"01";
    io_data_out_tb <= x"55";
    io_we_tb <= '0'; -- Write disabled
    wait until rising_edge(clk_tb);
    wait for 5 ns;
    assert port_out_a_tb = x"AA" -- Should still be previous value

```

```
        report "io_we='0' failed: Expected port unchanged (0xAA), got " &
            to_hstring(port_out_a_tb) severity error;

-- Test Case 3.1: IN from Port B
report "Test Case 3.1: IN from Port B";
io_addr_in_tb <= x"02";
pin_in_b_tb <= x"55";
wait for 10 ns; -- Allow combinational propagation
assert io_data_in_tb = x"55"
    report "IN from Port B failed: Expected 0x55, got " &
        to_hstring(io_data_in_tb) severity error;

-- Test Case 3.2: IN from Port A (read back)
report "Test Case 3.2: IN from Port A";
io_addr_in_tb <= x"01";
wait for 10 ns;
assert io_data_in_tb = x"AA"
    report "IN from Port A failed: Expected 0xAA, got " &
        to_hstring(io_data_in_tb) severity error;

-- Test Case 3.3: IN from undefined address
report "Test Case 3.3: IN from undefined address";
io_addr_in_tb <= x"00";
wait for 10 ns;
assert io_data_in_tb = x"00"
    report "Undefined address read failed: Expected 0x00, got " &
        to_hstring(io_data_in_tb) severity error;

-- Test Case 4.1: Output retention
report "Test Case 4.1: Output retention";
-- Write initial value
io_addr_in_tb <= x"01";
io_data_out_tb <= x"33";
io_we_tb <= '1';
wait until rising_edge(clk_tb);
io_we_tb <= '0';
wait for 5 ns;
assert port_out_a_tb = x"33"
    report "Initial write failed" severity error;

-- Wait multiple clock cycles
for i in 1 to 5 loop
    wait until rising_edge(clk_tb);
end loop;

-- Verify retention
assert port_out_a_tb = x"33"
    report "Retention failed: Expected 0x33, got " &
        to_hstring(port_out_a_tb) severity error;

-- Test Case 5.1: Reset after write
report "Test Case 5.1: Reset after write";
io_addr_in_tb <= x"01";
io_data_out_tb <= x"77";
io_we_tb <= '1';
wait until rising_edge(clk_tb);
io_we_tb <= '0';
wait for 5 ns;
assert port_out_a_tb = x"77"
    report "Pre-reset write failed" severity error;

reset_tb <= '1';
wait for 20 ns;
```



```

reset_tb <= '0';
wait for 10 ns;
assert port_out_a_tb = x"00"
    report "Reset failed: Expected 0x00, got " &
        to_hstring(port_out_a_tb) severity error;

-- Test Case 6.3: Simultaneous read/write
report "Test Case 6.3: Simultaneous read/write";
-- First write a value
io_addr_in_tb <= x"01";
io_data_out_tb <= x"99";
io_we_tb <= '1';
wait until rising_edge(clk_tb);
io_we_tb <= '0';
wait for 5 ns;

-- Read while setting up new write
io_addr_in_tb <= x"01"; -- Read from Port A
wait for 1 ns;
assert io_data_in_tb = x"99"
    report "Read before write failed" severity error;

-- Setup new write but don't clock yet
io_data_out_tb <= x"11";
io_we_tb <= '1';
-- Read should still show old value
wait for 1 ns;
assert io_data_in_tb = x"99"
    report "Read during write setup failed" severity error;

-- Execute write
wait until rising_edge(clk_tb);
io_we_tb <= '0';
wait for 5 ns;
assert port_out_a_tb = x"11"
    report "Write execution failed" severity error;

-- Now read should show new value
wait for 1 ns;
assert io_data_in_tb = x"11"
    report "Read after write failed" severity error;

report "All I/O Unit test cases completed successfully!";
wait;
end process;

-- Monitor process for debugging
monitor_proc: process
begin
    wait on io_addr_in_tb, pin_in_b_tb, io_we_tb, clk_tb;
    wait for 1 ns; -- Allow for propagation
    report "Time: " & time'image(now) &
        " | Addr: " & to_hstring(io_addr_in_tb) &
        " | io_we: " & std_logic'image(io_we_tb) &
        " | DataOut: " & to_hstring(io_data_out_tb) &
        " | DataIn: " & to_hstring(io_data_in_tb) &
        " | PortA: " & to_hstring(port_out_a_tb) &
        " | PinB: " & to_hstring(pin_in_b_tb);
end process;

end Behavioral;

```

4.2 Test Sequence

1. **Initialization:** Apply reset, verify ports initialized to zero
2. **OUT Instruction Tests:** Verify write operations to output port
3. **Address Selection Tests:** Verify correct port addressing
4. **IN Instruction Tests:** Verify read operations from input pins
5. **Output Retention Tests:** Verify values persist until changed
6. **Reset Tests:** Verify reset clears output registers
7. **Timing Tests:** Verify setup/hold times and propagation delays
8. **Integration Tests:** Test read/write combinations and sequences
9. **Corner Cases:** Test boundary addresses, all-zero/all-one patterns

4.3 Monitoring and Logging

- Continuous monitoring of all I/O signals
- Log all read/write operations with timestamps
- Track output port values over time
- Capture timing diagrams for critical operations
- Generate I/O transaction log for analysis

5 PASS/FAIL CRITERIA

5.1 Success Criteria

Criteria	Description
Functional	All test cases execute without assertion failures
Timing	Writes on rising clock edge; reads immediate (combinational)
Coverage	All 5 requirements fully verified
Addressing	Correct port selection for all valid addresses
Retention	Output values maintained until explicitly changed
Reset	Complete and reliable clearing of output registers

5.2 Failure Conditions

- Write doesn't occur when `io_we='1'` and address valid
- Read returns incorrect value for given address
- Output port loses value without explicit write or reset
- Reset doesn't clear output port to zero
- Address decoding incorrect (writes to wrong or undefined ports)
- Timing violations in read/write operations

5.3 Coverage Goals

- **Requirement Coverage:** 100% of I/O requirements (5/5)
- **Address Coverage:** All 256 addresses sampled for read behavior
- **Data Pattern Coverage:** Key patterns tested (00, FF, 55, AA, etc.)
- **Timing Coverage:** Setup/hold times, propagation delays verified
- **Sequence Coverage:** Common I/O sequences (IN-OUT combinations)

6 SPECIAL TEST CONSIDERATIONS

6.1 Asynchronous Read Behavior

- Verify io_data_in updates immediately with address/pin changes
- Test reads without clock dependency
- Measure propagation delay from input change to data valid
- Check for glitches during address or input transitions

6.2 Synchronous Write Behavior

- Verify writes occur only on rising clock edge with io_we='1'
- Test setup and hold time requirements
- Verify write data sampled at clock edge, not before/after
- Test maximum write frequency

6.3 Reset Characteristics

- Verify reset is asynchronous (takes effect immediately)
- Test reset pulse width requirements
- Verify reset overrides write operations
- Test recovery from reset (ability to write after reset)

6.4 Address Space Management

- Verify only address 0x01 maps to output port
- Test that writes to other addresses have no effect
- Verify reads from undefined addresses return 0x00
- Check address decoding doesn't create bus conflicts

6.5 Integration with CPU

- Verify signal compatibility with Control Unit outputs
- Test interface with Register File for data transfer
- Validate timing with instruction execution cycles
- Check for proper handshake with CPU pipeline

6.6 Physical Interface Considerations

- Test behavior with floating inputs (should read as defined)
- Verify output drive characteristics (simulation)
- Check for output glitches during transitions
- Test with realistic load conditions on output pins

6.7 Error Conditions

- Test behavior with metastable inputs (simulation only)
- Verify recovery from invalid address patterns
- Check for X-propagation in simulation
- Test power-on behavior without explicit reset

7 SIGN-OFF

7.1 Test Results Summary

Category	Metric	Target	Achieved
Requirements	Requirements Covered	5/5	
Test Cases	Test Cases Executed	35/35	
Coverage	Requirement Coverage	100%	
Coverage	Address Space Coverage	100%	
Coverage	Data Pattern Coverage	100%	
Quality	Assertion Pass Rate	100%	
Timing	Write Setup Time	Meets spec	
Timing	Read Propagation Delay	<5ns	

7.2 Approval Signatures

Role	Name	Signature	Date
Test Engineer			
Lead Engineer			
Quality Assurance			
Project Manager			

7.3 Test Closure

- All 35 test cases executed and verified
- 100% requirement coverage achieved
- OUT instruction correctly writes to output port (TR-IO-01)
- Correct port selection based on address verified (TR-IO-02)
- IN instruction correctly reads input pins verified (TR-IO-03)
- Output retention until update/reset confirmed (TR-IO-04)
- Reset clears output ports to zero verified (TR-IO-05)
- All timing requirements met
- No address decoding errors or bus conflicts
- All test artifacts archived per configuration management procedures
- I/O Unit ready for integration with CPU and external devices